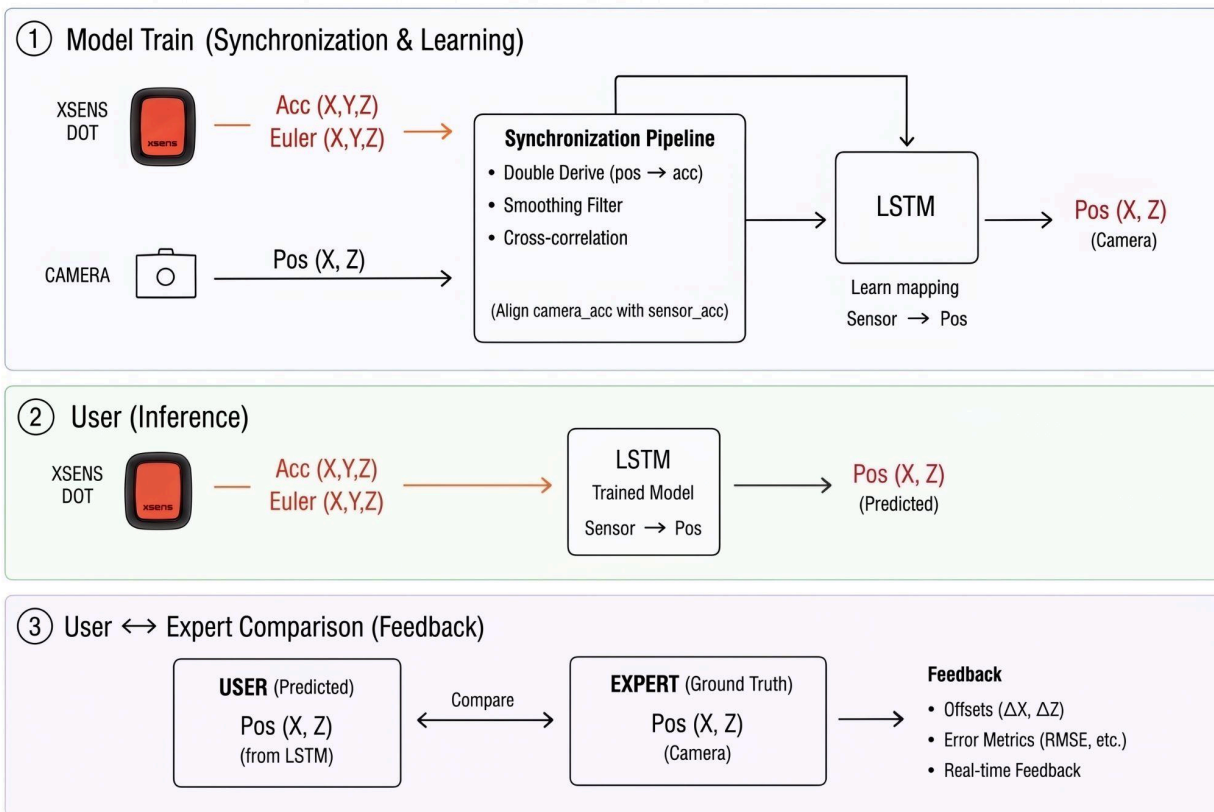


Project Summary: Exercise Correction Using Wearable Devices

Role: Machine Learning Engineer

Project Overview¹



We developed a sophisticated software and hardware ecosystem to provide real-time exercise correction. The project operates on a three-step machine learning architecture:

- 1. Model Train (Synchronization & Learning):** The LSTM model is trained using synchronized data from both an XSENS DOT IMU wearable sensor (acceleration and Euler angles) and a camera (positional data X, Z). The synchronization pipeline aligns the data by double-deriving the camera's position to acceleration, applying a smoothing filter, and using cross-correlation. The camera's positional data serves as the definitive answer labels for the LSTM to learn the sensor-to-position mapping.
- 2. User (Inference):** Once the mapping is learned, users perform exercises using only the XSENS DOT wearable sensor. The trained LSTM model takes the real-time sensor data and outputs the predicted position (X, Z) without requiring any camera input.
- 3. User ↔ Expert Comparison (Feedback):** The user's predicted positional data is compared directly against an expert's ground truth positional data. By

¹ <https://github.com/JeongjunBae/Exercise-correction-using-wearable-devices/blob/main/README-EN.md>

calculating positional offsets (ΔX , ΔZ) and error metrics (such as RMSE), the system provides actionable, real-time posture feedback to the user.

For a visual and detailed reference regarding the project's background and development, please refer to the presentation slides². Please note that a band-pass filter may be used for some exercises.

Performance Metrics:

Through this pipeline, the system achieved high precision in tracking, resulting in:

- Vertical Error: **2.5cm**
- Horizontal Error: **4cm**

Background & Motivation

- Although wearable-device-based motion tracking has been actively studied, **existing approaches often require numerous sensors**, making them impractical for everyday users to wear during exercise.
- The **high cost of personal training** creates a significant barrier to accessible and proper posture correction.
- Muscle imbalances are highly common. Individuals naturally default to unbalanced and potentially harmful postures when exhausted during intense weight training.

Technical Implementation Overview

Component	Technology / Approach	Details
IMU Sensors	Xsens Dot	Utilized a highly accurate MEMS sensor for raw data gathering.
Vertical Tracking	Band-pass & High-pass filters	Mitigated integration drift by isolating the vertical position through sequential filtering.
Horizontal Tracking	LSTM Neural Network	Mapped horizontal acceleration directly to camera tracking data to avoid exponential drift.

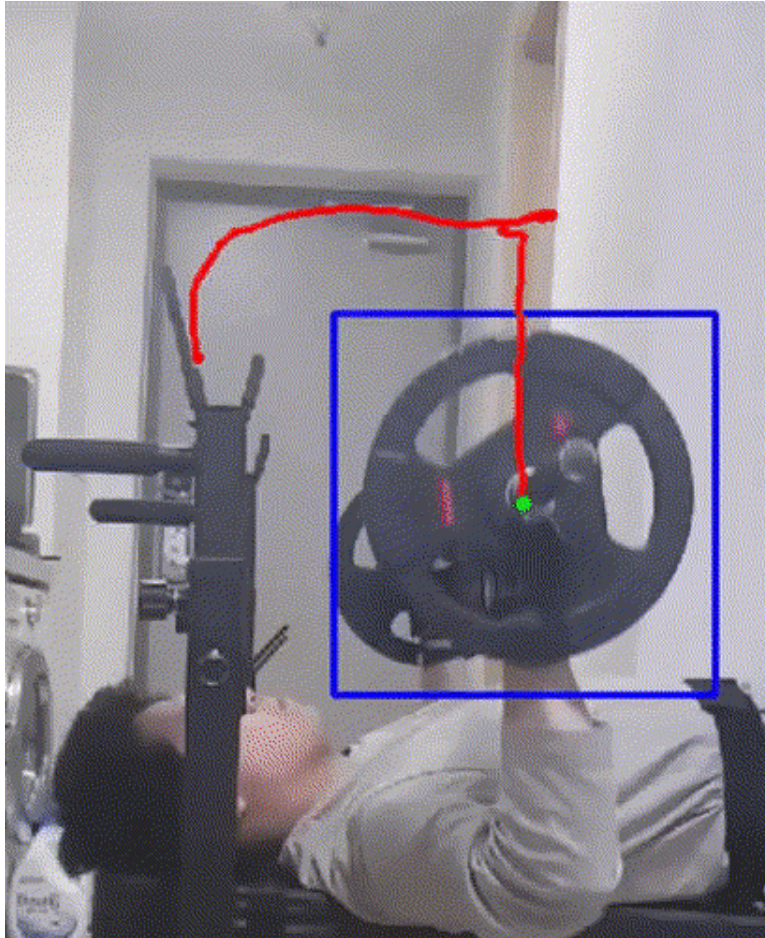
Algorithms & Data Processing

- **Missing Data Handling:** Missing data points caused by sensor transmission drops were handled by utilizing intersecting timeframes and applying linear interpolation.
- **Synchronization³:** Data synchronization between the IMU sensor and the camera was achieved using Fourier Transforms.

² https://github.com/JeongjunBae/Exercise-correction-using-wearable-devices/blob/main/final_presentation_English.pdf

³ https://github.com/JeongjunBae/Synchronize_Signals/blob/main/ReadMe-EN.md

- **Computer Vision:** The pipeline sets a physical scale based on the barbell's size and precisely tracks the center of the weight plates.



- **Exercise Evaluation:** The algorithm detects signal peaks to compare maximum heights and reach times. It further evaluates the exercise cycle by measuring the time between peaks and calculates standard deviations to ensure cycle consistency.

Challenges

- Mapping acceleration directly to position yielded less-than-ideal results initially, likely due to the differing physical properties of the metrics.
- The third-party WL-Analysis app⁴ failed to recognize several exercises, forcing us to quickly develop our own custom tracking program.

Future Work

- Design more robust algorithms to further minimize horizontal positional errors.
- Increase the system's capabilities beyond the currently supported exercises (Military Press, Squat, Barbell Row, and Deadlift).

Awards

- **First Prize:** 2021 Kyung Hee University Creative Design Competition.⁵
- **Second Prize:** 2021 Sungkyunkwan University (SKKU) Creative Design Competition.⁶

⁴ https://play.google.com/store/apps/details?id=com.karolsmolak.wlanalysis&hl=en_US&pli=1

⁵ https://drive.google.com/file/d/11RK3T1I0vtgmYwRjneoGL2eivUWNryM/view?usp=drive_link

⁶ https://drive.google.com/file/d/1ITSIbwsLu7ajZHV0UYx7xbmm7YR7AM7f/view?usp=drive_link